Metacommunities I

Today's Agenda:

- Quiz
- Metacommunities

Patchy Environments

Patches: suitable habitat, surrounded by unsuitable habitat.

Metapopulation: collection of connected populations

- Multiple patches connected by dispersal

Patchy Environments

Patches: suitable habitat, surrounded by unsuitable habitat.

Metapopulation: collection of connected populations

- Multiple patches connected by dispersal

Metacommunity: collection of connected communities

- Multiple patches connected by dispersal of one or more species

Metacommunities

4 Main perspectives:

1) Patch dynamics - extension of metapopulation stuff to more species
2) Species sorting - community assembly-ish, species differ in abiotic niches
3) Mass effects - species sorting-ish, w/ source-sink and rescue effects
4) Neutral theory - assumes species are identical, environment irrelevant

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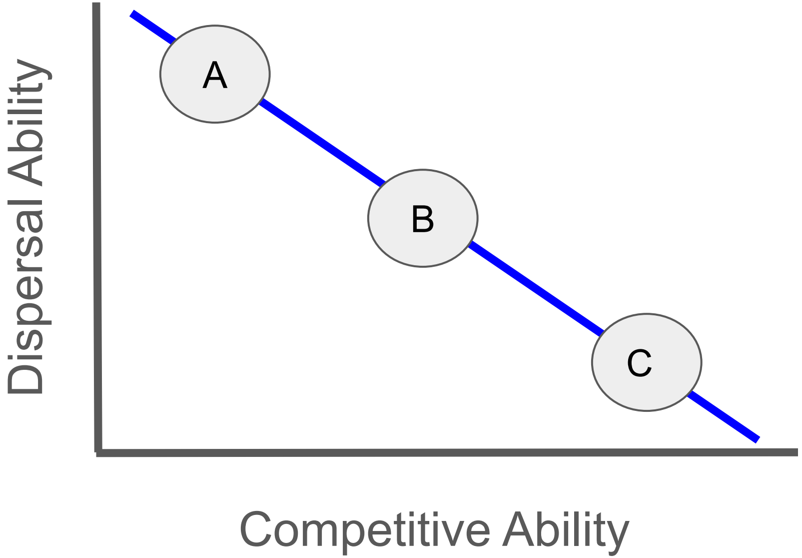
Patch Dynamics Perspective

- This is what we covered last week
- Metacommunity studies focus on multiple species, and hence coexistence
  - Various trade-offs and coexistence mechanisms



1) A arrives first
2) B arrives next, displaces A
3) C arrives last, displaces B
4) Patch extinction occurs (go to 1)

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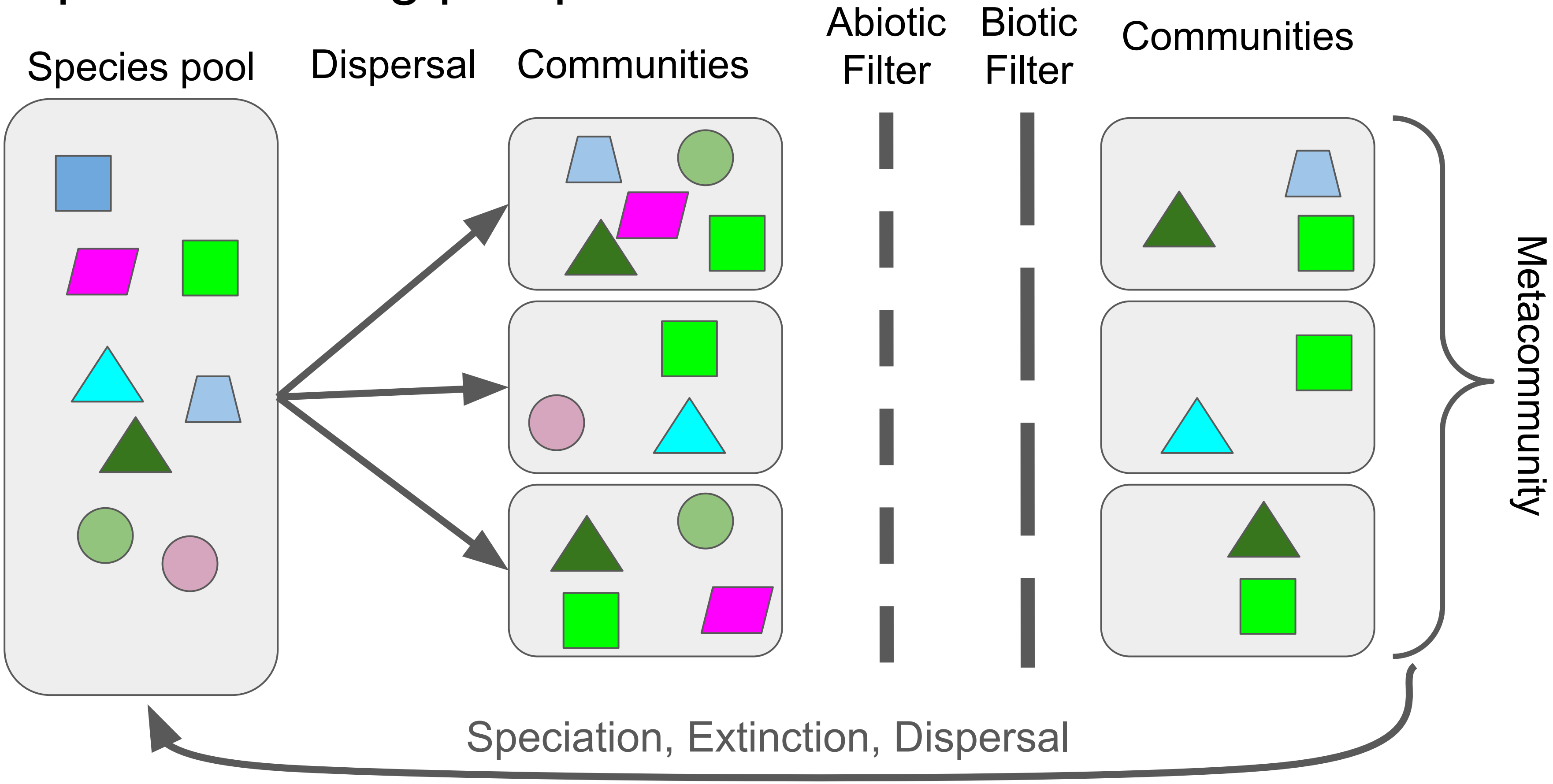
Species sorting perspective

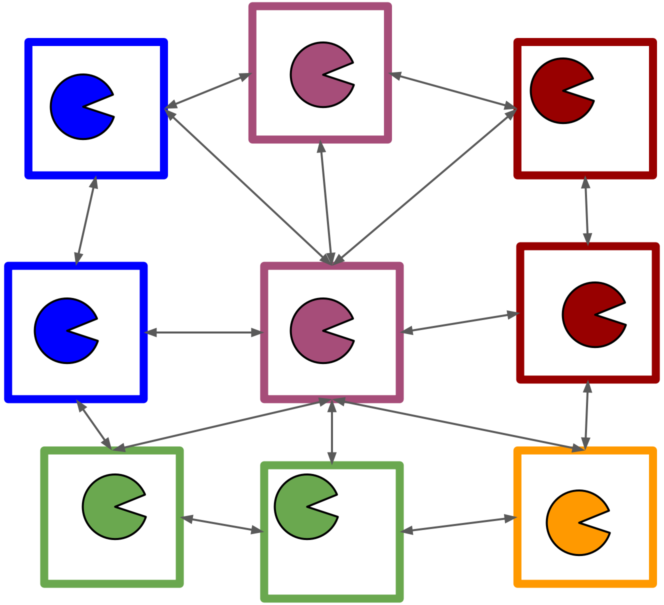
Species sorting perspective

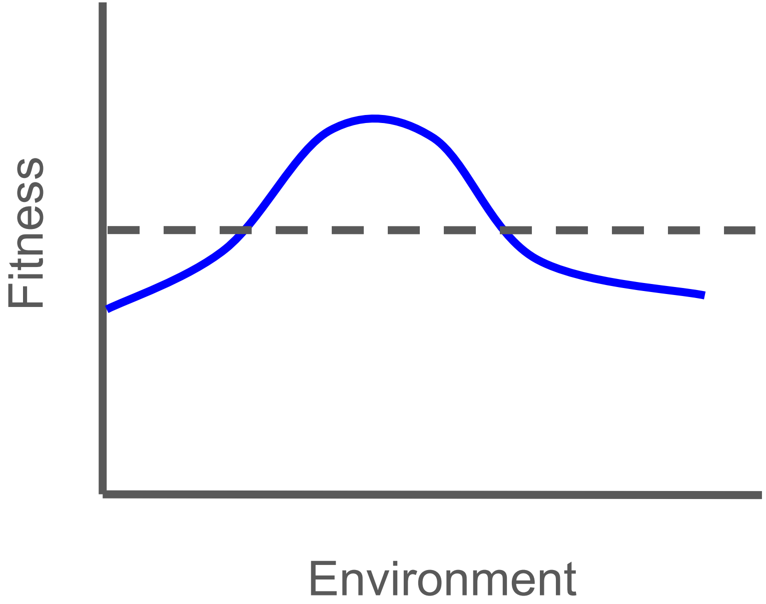


Species sorting perspective

- Species performance varies with environment
- Dispersal is high enough for species to reach all patches
- Dispersal low enough that it doesn’t override performance differences

Sorting perspective





Outside

Species sorting perspective

- Repeated dispersal and filtering allows tracking of changes
    - E.g., shifts due to changing climate
- Allows abiotic filters to shift overtime, or even interact

Species sorting perspective

Allows us to partition variation in community membership/abundance

- How much beta diversity is due to environment?
- How much is due to dispersal?

Species sorting perspective

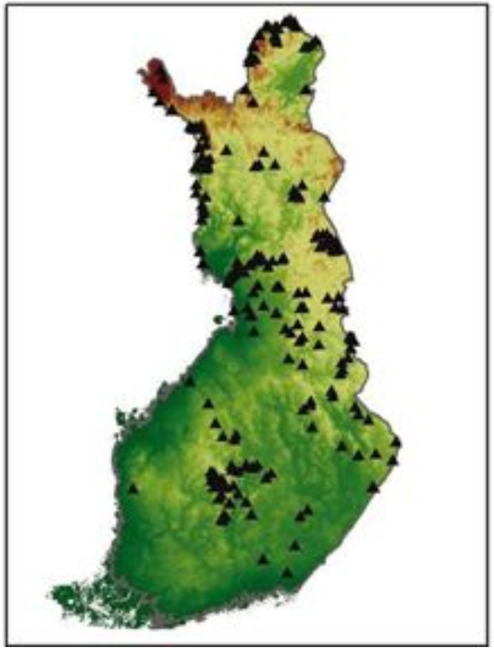
- Diatoms in Finland

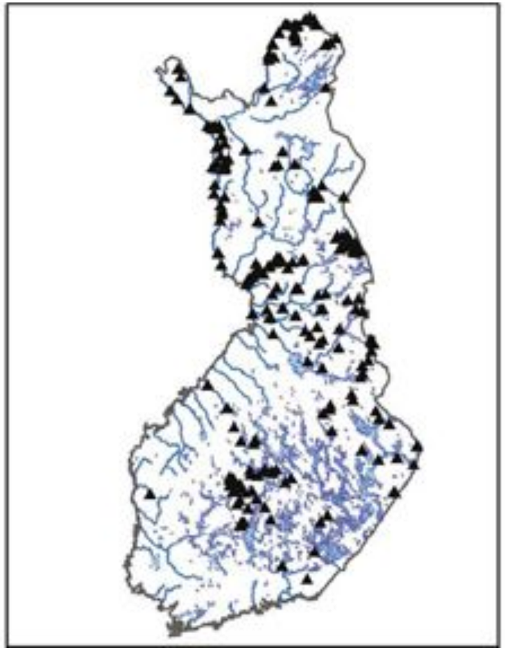
Oikos 119: 129–137, 2010
doi: 10.1111/j.1600-0706.2009.17778.x,

Subject Editor: Daniel Gruner. Accepted 8 June 2009

Geographical patterns of micro-organismal community structure: are diatoms ubiquitously distributed across boreal streams?

Jani Heino, Luis Mauricio Bini, Satu Maaria Karjalainen, Heikki Mykrä, Janne Soininen, Ludgero Cardoso Galli Vieira and José Alexandre Felizola Diniz-Filho

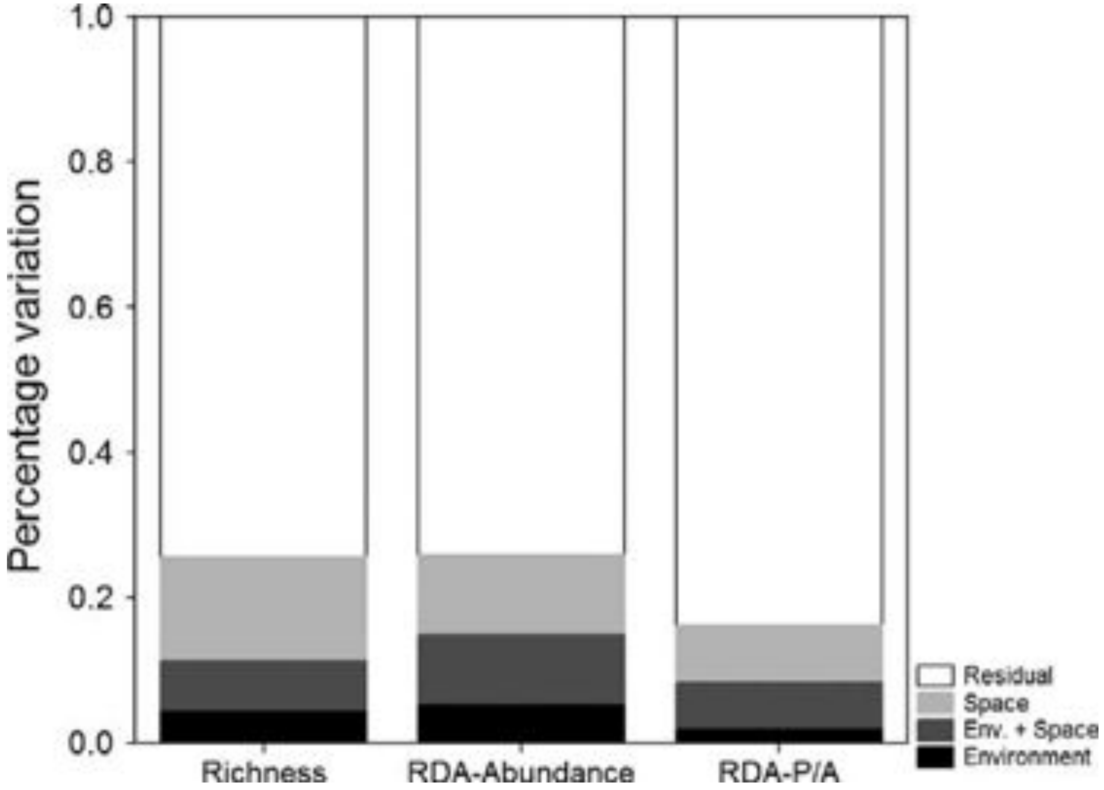




Variable	Mean	SD	Minimum	Maximum
Altitude (m a.s.l.)	151	77.96	4	539
Catchment area (km^2)	2357	5990	7	40131
Peatland cover (%)	28	17.34	0	75
pH	6.8	0.52	4.9	7.8
Conductivity (mS m^{-1})	3.95	1.99	1.5	18.0
Colour (mg Pt l^{-1})	79	69.31	3	400
Total-P (µg l^{-1})	13.46	8.98	1	52
Total-N (µg l^{-1})	335	173	50	1035

488 species of diatom

Species sorting perspective



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Species sorting perspective

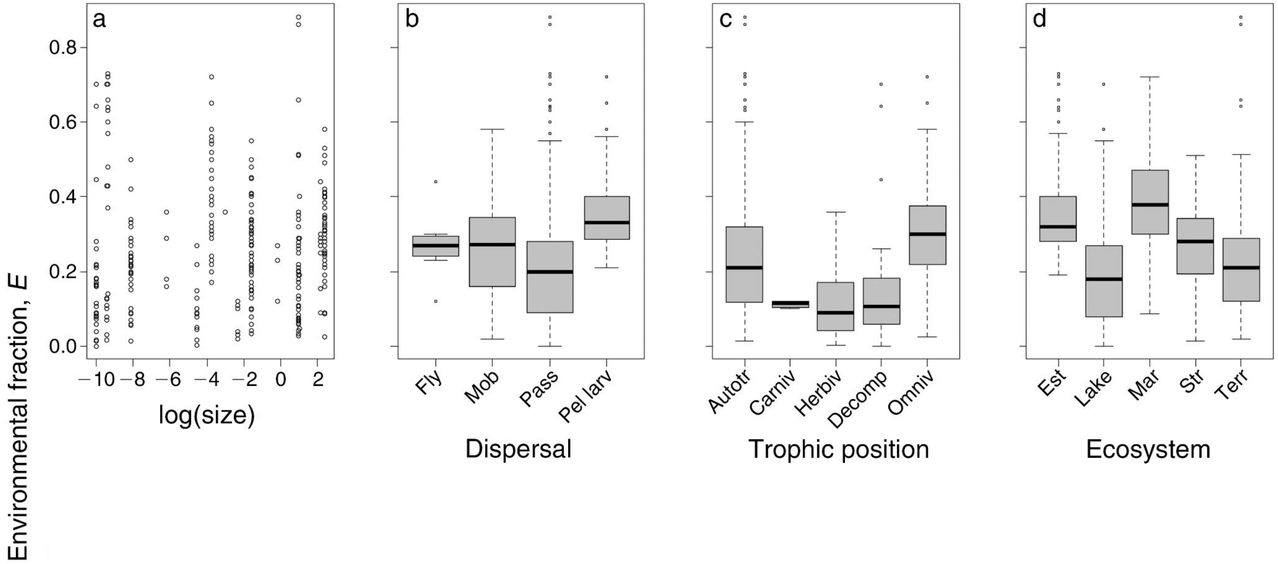
Ecology, 95(12), 2014, pp. 3284–3292

A quantitative analysis of species sorting across organisms and ecosystems

JANNE SOININEN[1]

Meta-analysis

Only 26% of variation attributable to environment



Species sorting perspective

The relative importance of environment is still an open question

- More mechanistic, trait-based perspective may help

Metacommunities

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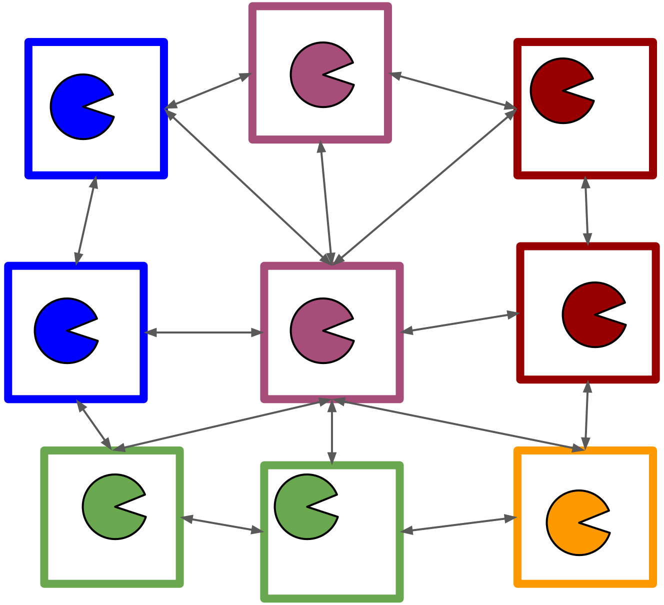
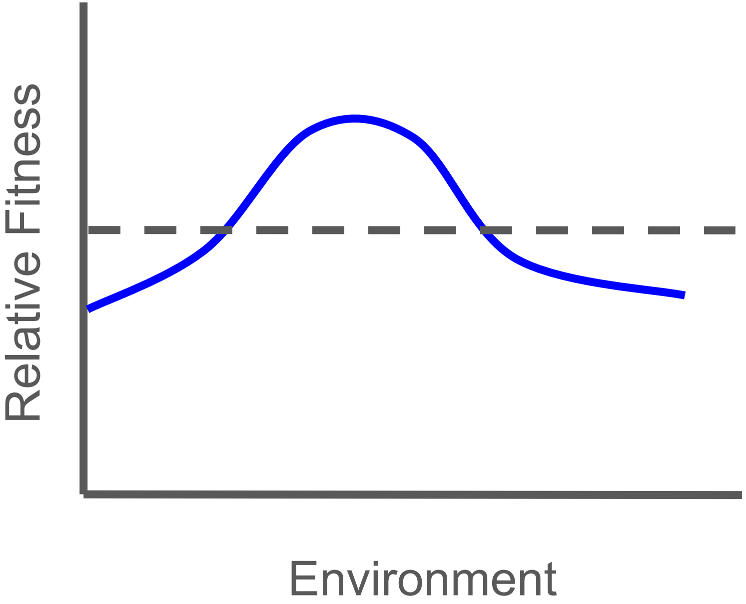
Mass-effects

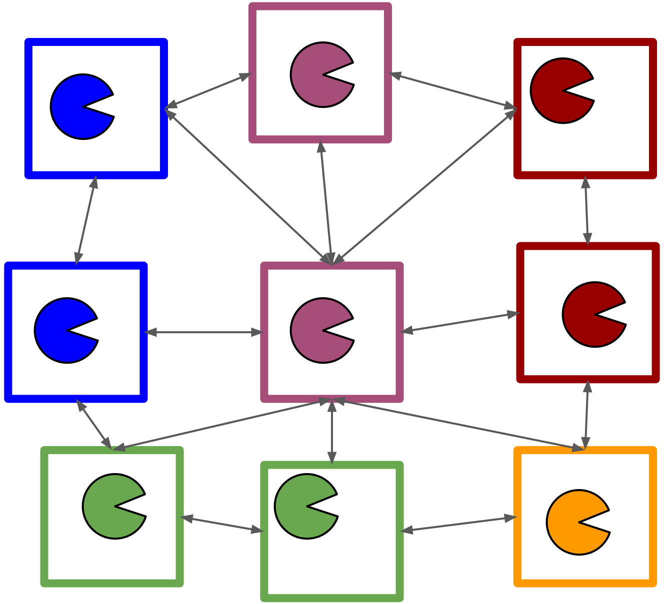
- Similar to species sorting
  - Species have different niches
  - Assumes patches will differ
  - Species do well/poorly in different patches based on niches and environment
- Differences
  - High-quality patches can serve as “sources”
  - Low-quality patches can serve as “sinks”

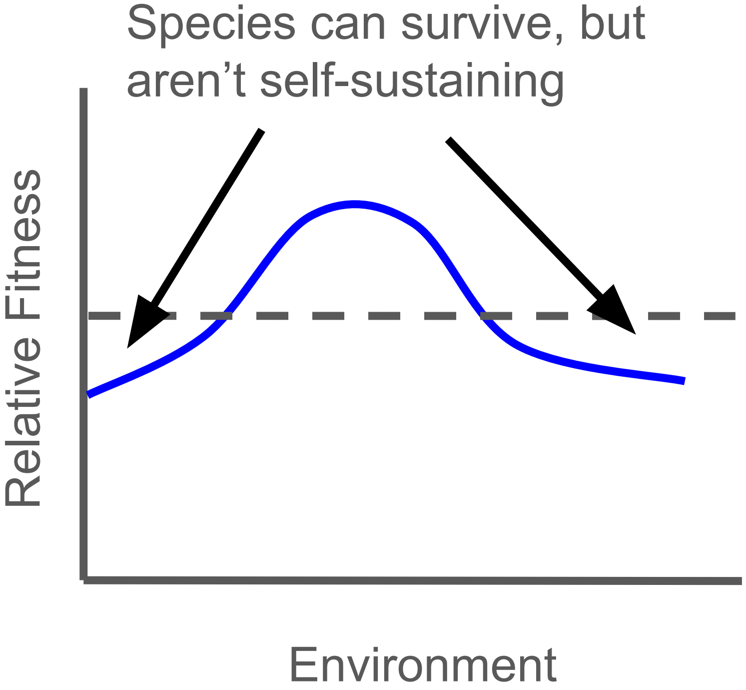
Sorting perspective



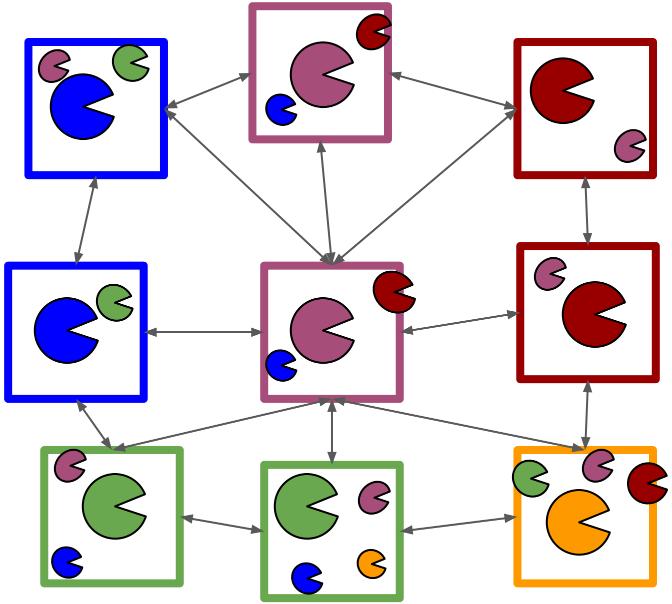


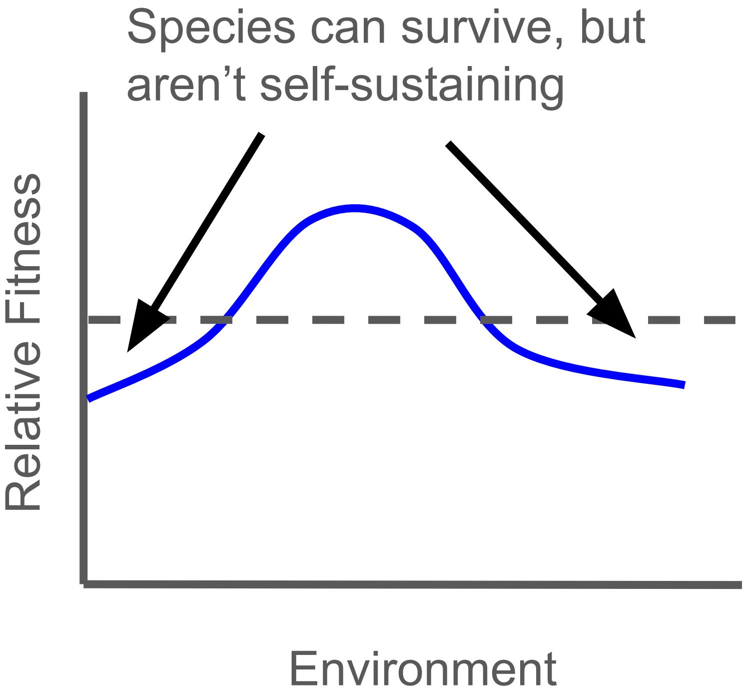
Mass effect




Species can survive, but aren’t self-sustaining
Relative Fitness
Environment

Mass effect





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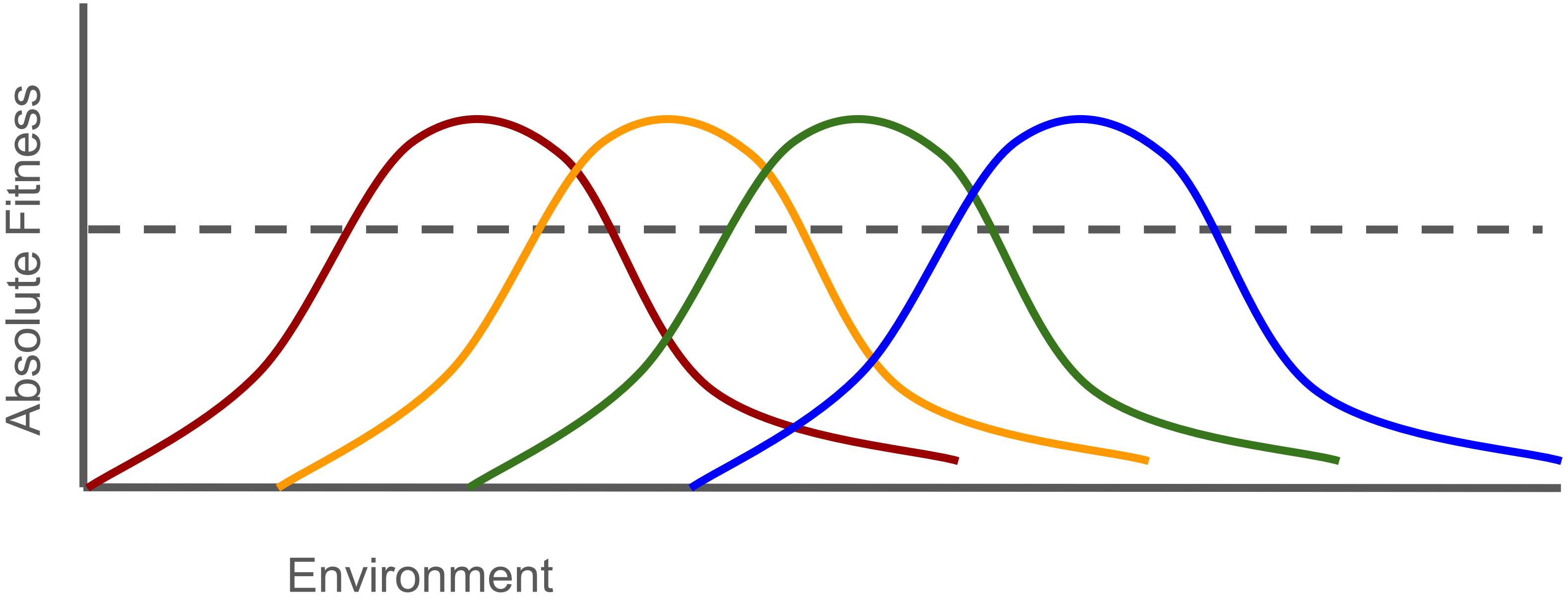
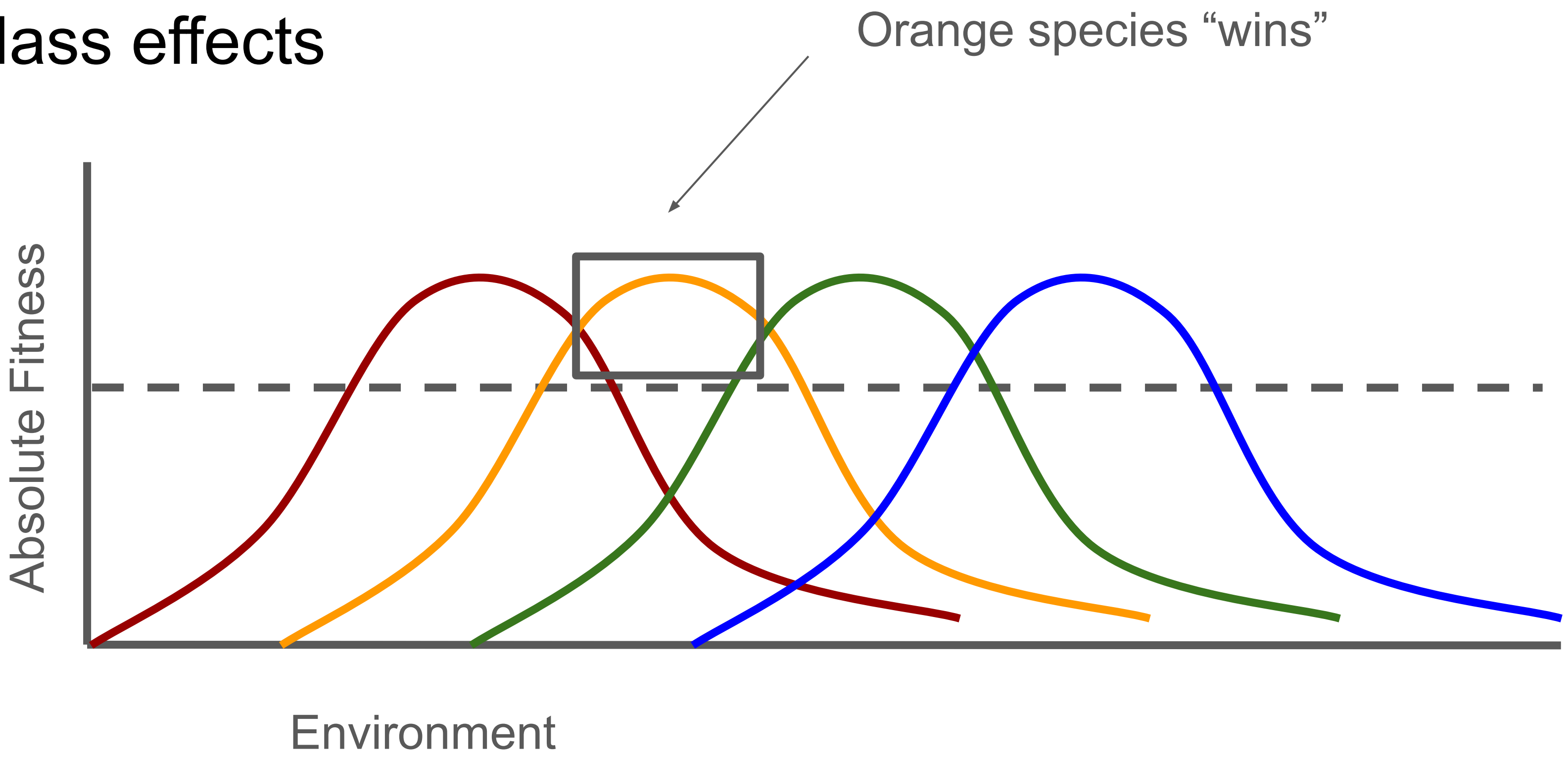
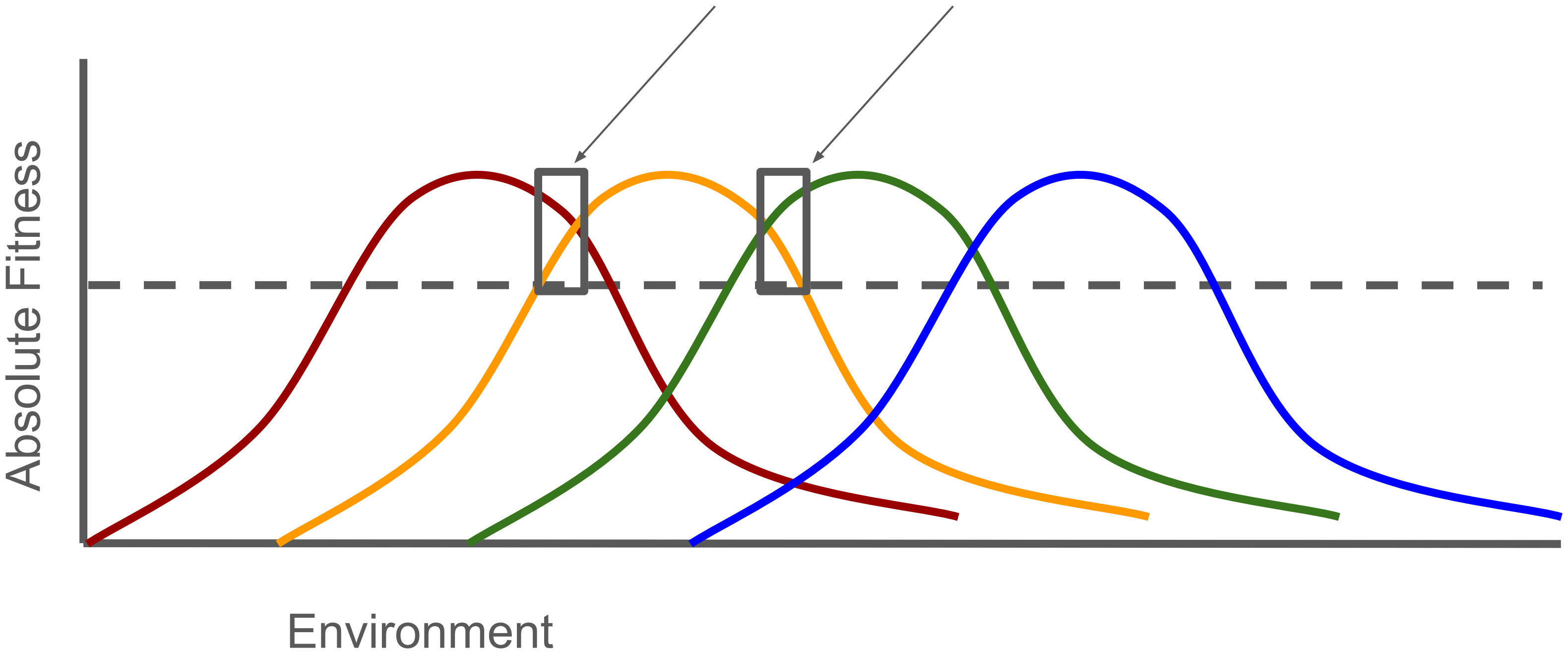


Mass effects



Mass effects

Orange species "loses", but can persist due if immigration occurs



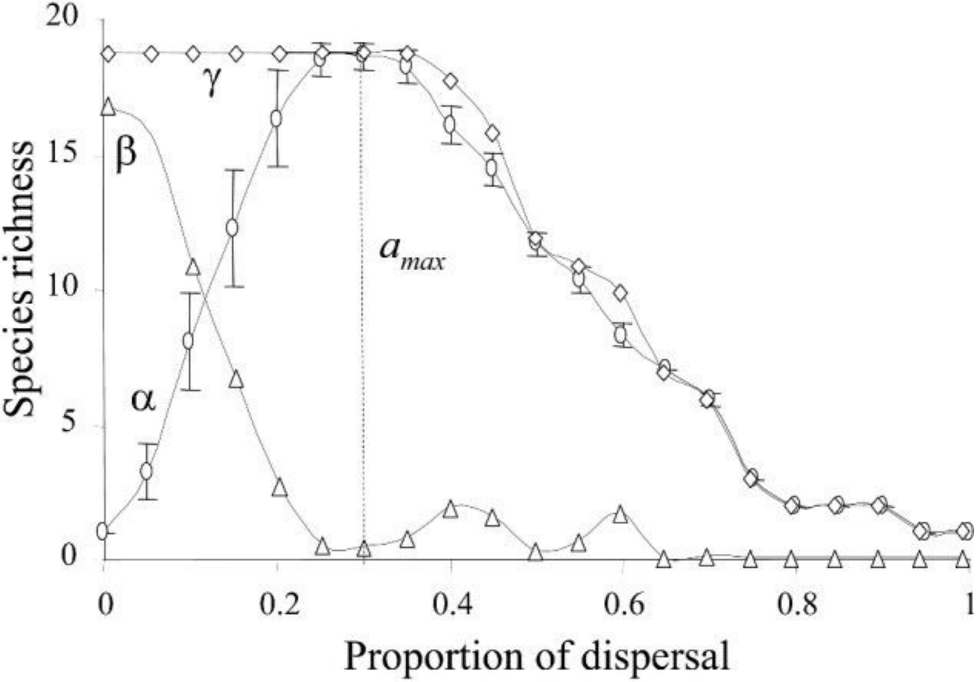
Mass effect simulation


VOL. 162, NO. 5 THE AMERICAN NATURALIST NOVEMBER 2003

Community Patterns in Source-Sink Metacommunities

Nicolas Mouquet[1,*] and Michel Loreau[2]

- Species differ in niches, but have identical dispersal
- Varied dispersal and studied diversity effects



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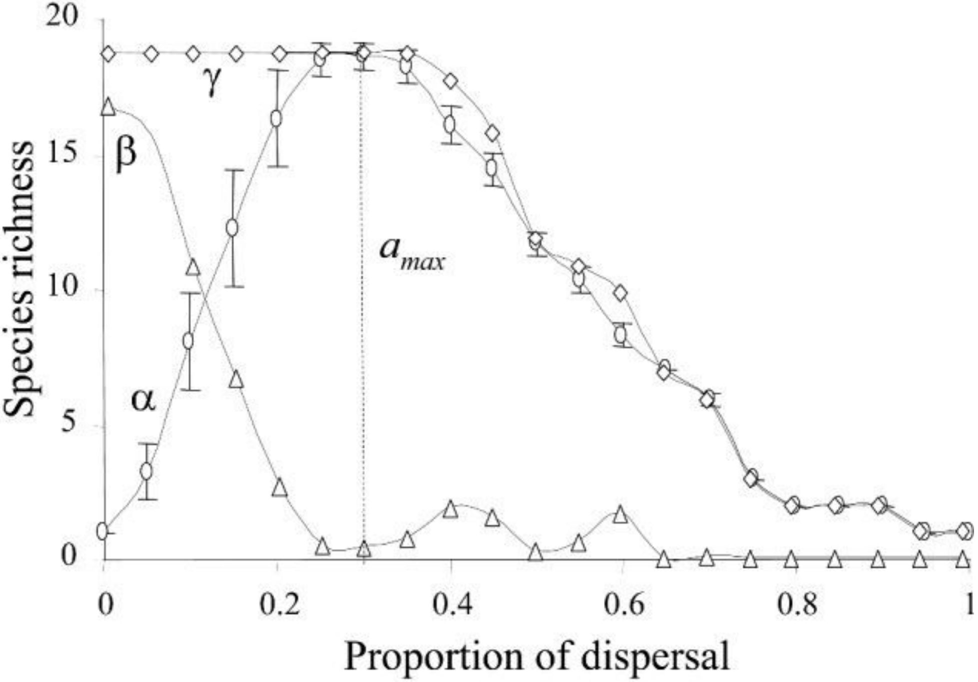

No dispersal:
one species dominates each patch

Lots of dispersal:
One species dominates all patches

Mass effect simulation

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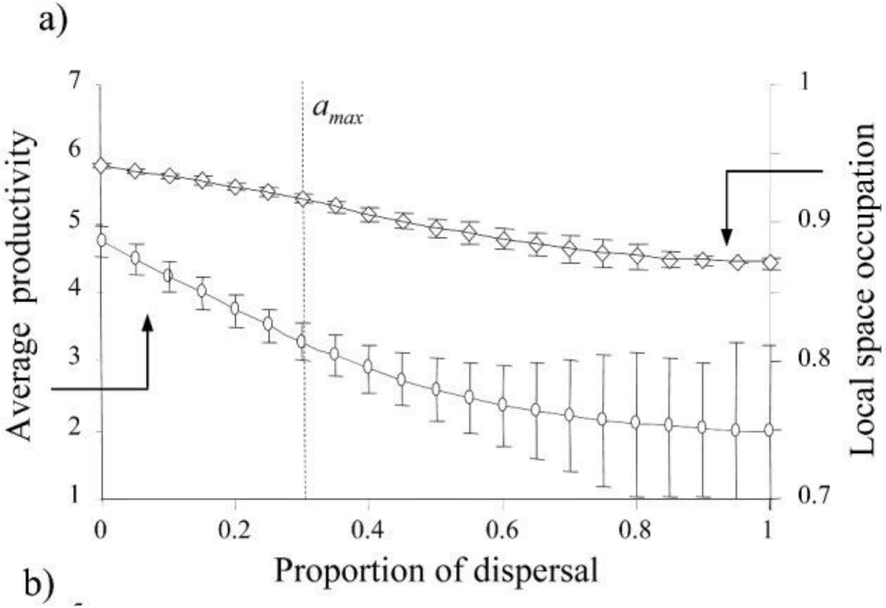

Empirical mass effect work

- Some field research that examines this
  - E.g., different connectivity due to dispersal mode or patch connectivity

- Some cool experiments that manipulate dispersal between patches
  - E.g., mesocosms that are connected for varying lengths of time

Mass effect in the field

Oikos 116: 1255–1266, 2007
doi: 10.1111/j.2007.0030-1299.15860.x,

Subject Editor: Veijo Kaitala, Accepted 20 March 2007

Research

The role of metacommunity processes in shaping invertebrate rock pool communities along a dispersal gradient

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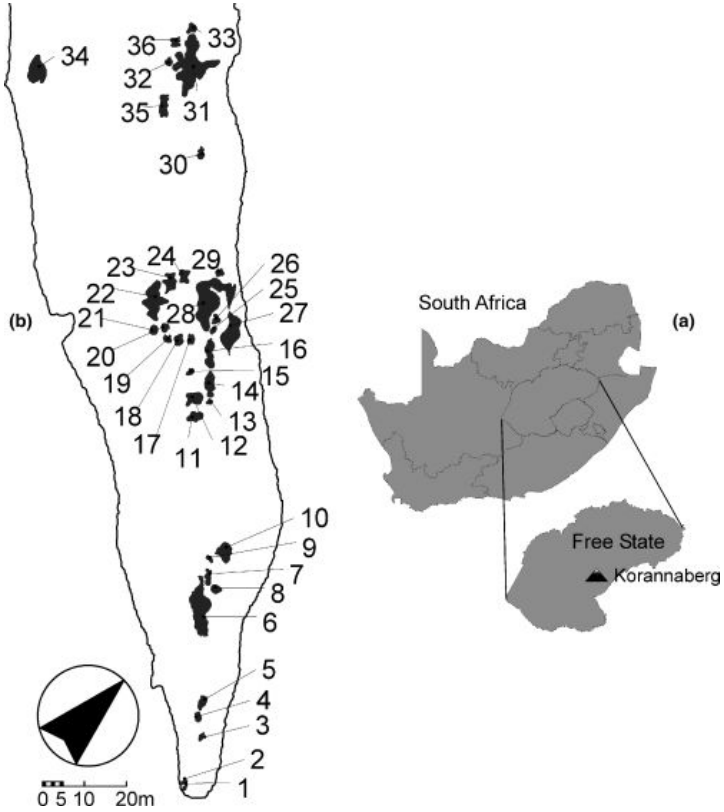
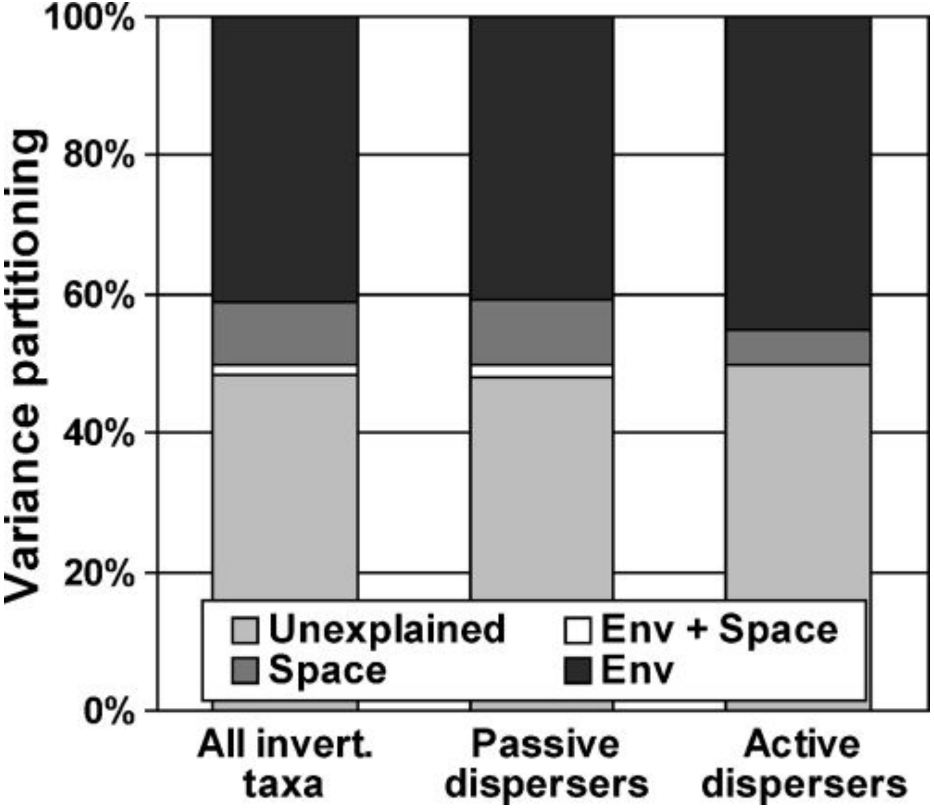
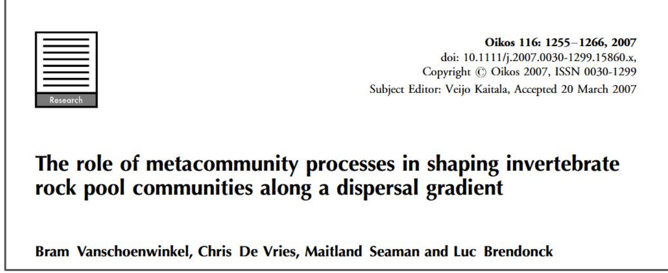


- Sampled invertebrates in temporary pools
- Categorized as active vs passive dispersers

Mass effect in the field

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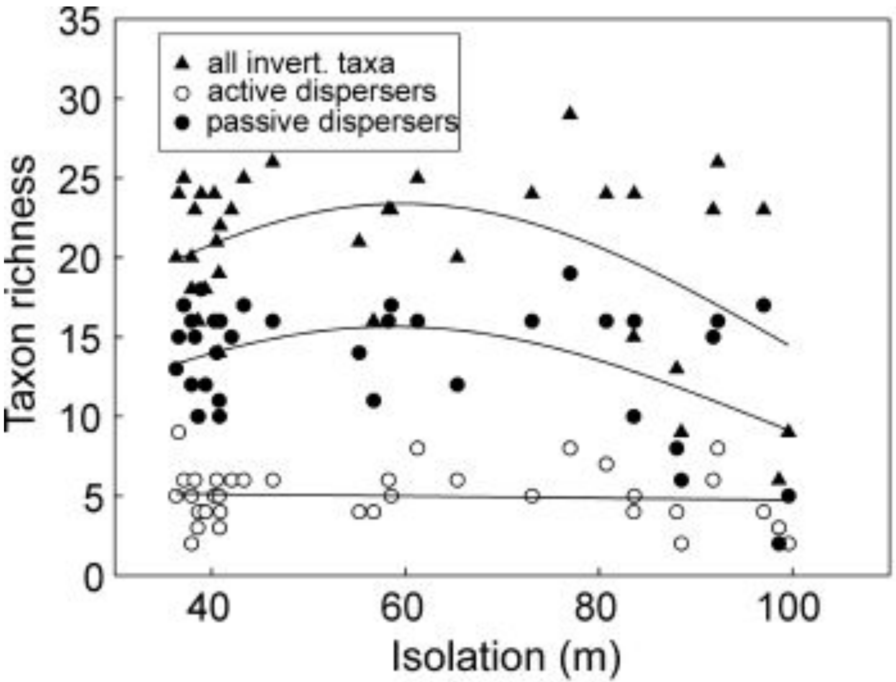

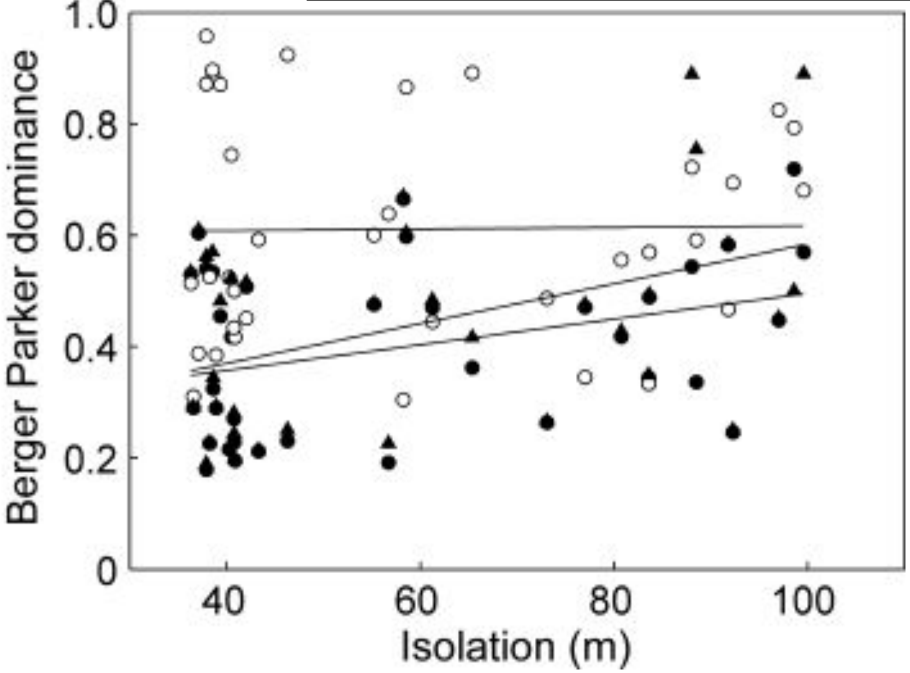

Mass effect experiments

- Focused on communities in pitcher plants
- Manipulated:
  - resources (dead ants)
  - Predation (mosquito larvae)
  - dispersal

VOL. 162, NO. 2 THE AMERICAN NATURALIST AUGUST 2003

Dispersal Rates Affect Species Composition in Metacommunities of *Sarracenia purpurea* Inquilines

Jamie M. Kneitel* and Thomas E. Miller†

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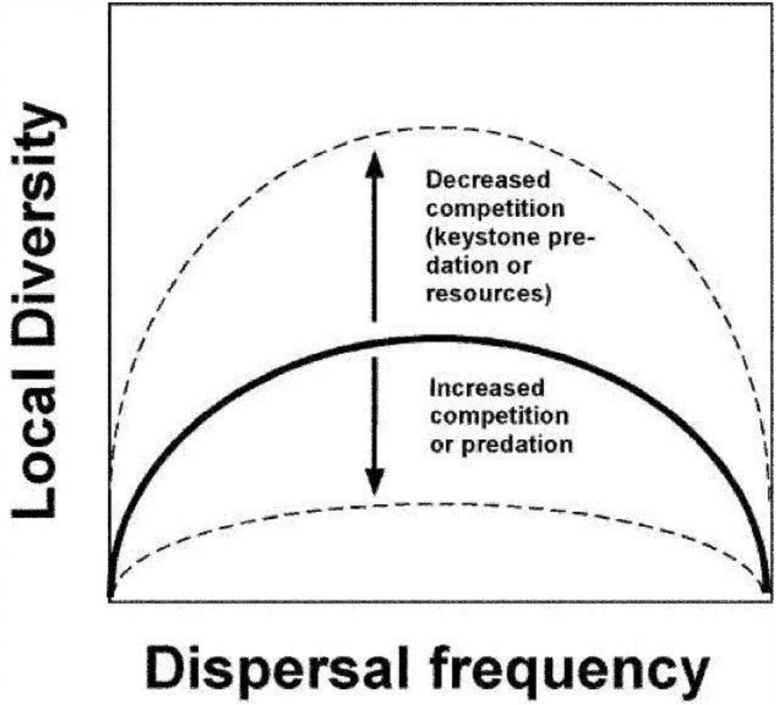

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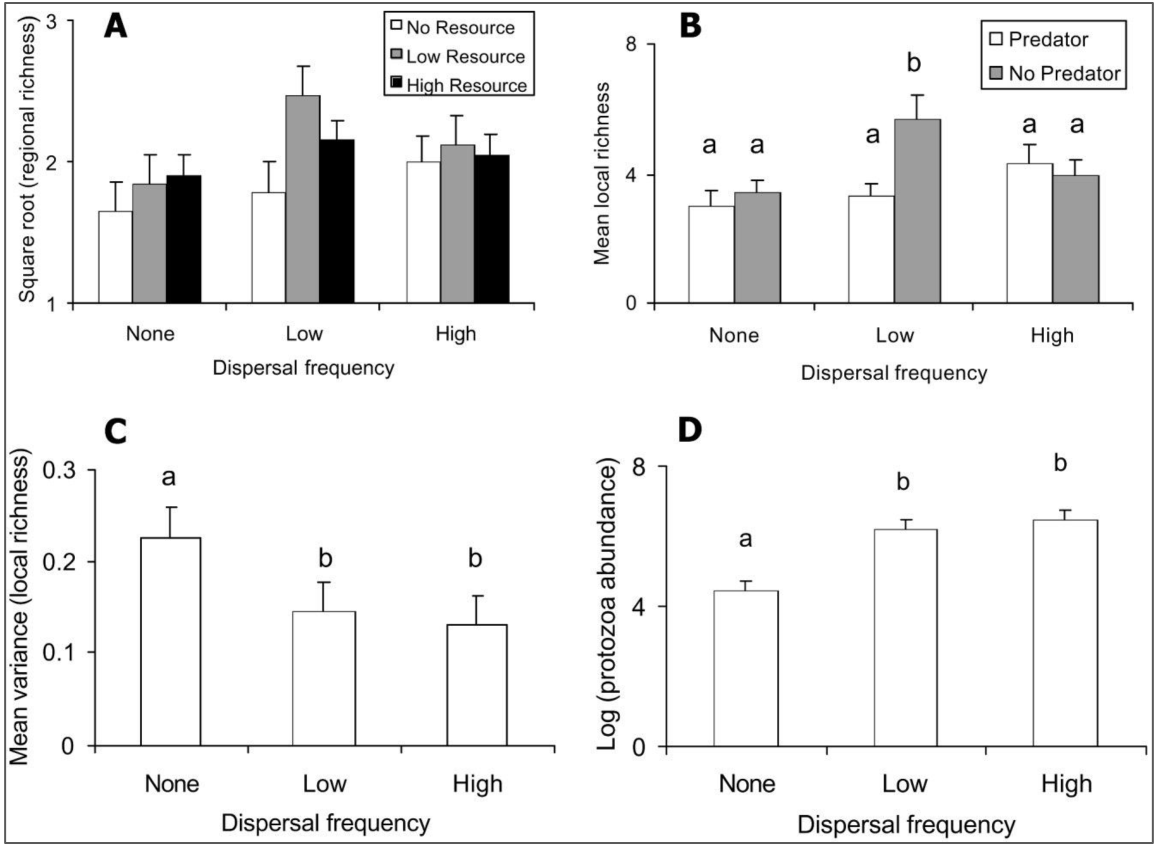

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Next time!

Next class: More metacommunities